

```
# Step 1: Import required libraries

import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, MinMaxScaler


# Step 2: Dataset source and description

csv_url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'

print("Dataset: Iris Dataset")
print("Source: UCI Machine Learning Repository")
print("URL:", csv_url)

print("\nDescription:")
print("The dataset contains 150 samples of iris flowers.")
print("Each sample has 4 features: sepal length, sepal width, petal length, petal width.")
print("Target variable: species (Setosa, Versicolor, Virginica)")


# Step 3: Load dataset

column_names = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']

df = pd.read_csv(csv_url, header=None, names=column_names)

print("\nDataset loaded successfully!\n")
print(df.head())
```

Step 4: Data Preprocessing

```
print("\n--- Data Preprocessing ---")
```

Missing values

```
print("\nMissing Values:")
```

```
print(df.isnull().sum())
```

Statistical summary

```
print("\nStatistical Summary:")
```

```
print(df.describe())
```

Dataset dimensions

```
print("\nShape of dataset:")
```

```
print(df.shape)
```

Data types and info

```
print("\nDataset Info:")
```

```
print(df.info())
```

Variable description

```
print("\nVariable Details:")
```

```
for col in df.columns:
```

```
    if df[col].dtype == 'object':
```

```
        print(f"{col} → Categorical")
```

```
    else:
```

```
        print(f"{col} → Numerical (Min: {df[col].min()}, Max: {df[col].max()})")
```

Step 5: Data Formatting & Normalization

```
print("\n--- Data Formatting ---")
```

```
# Convert species to category (factor type)
```

```
df['species'] = df['species'].astype('category')
```

```
print("\nUpdated Data Types:")
```

```
print(df.dtypes)
```

```
# Normalization using Min-Max Scaling
```

```
print("\n--- Data Normalization ---")
```

```
print("Formula: (x - min) / (max - min)")
```

```
scaler = MinMaxScaler()
```

```
df[['sepal_length']] = scaler.fit_transform(df[['sepal_length']])
```

```
print("\nNormalized 'sepal_length':")
```

```
print(df[['sepal_length']].head())
```

Step 6: Convert categorical variables to numerical

```
print("\n--- Encoding Categorical Data ---")
```

```
encoder = LabelEncoder()
```

```
df['species_encoded'] = encoder.fit_transform(df['species'])
```

```
print("\nOriginal Species:")
```

```
print(df['species'].unique())
```

```
print("\nEncoded Values:")
```

```
print(df['species_encoded'].unique())
```

```
# Mapping
```

```
mapping = dict(zip(encoder.classes_, encoder.transform(encoder.classes_)))
```

```
print("\nEncoding Map:")
```

```
print(mapping)
```

```
print("\nFinal Dataset:")
```

```
print(df.head())
```